

PICO CTF- CHALLENGE 1

Riddle Registry 

Easy Forensics picoMini by CMU-Africa browser_webshell_solvable

AUTHOR: PRINCE NIYONSHUTI N.

Description

Hi, intrepid investigator!  You've stumbled upon a peculiar PDF filled with what seems like nothing more than garbled nonsense. But beware! Not everything is as it appears. Amidst the chaos lies a hidden treasure—an elusive flag waiting to be uncovered.
Find the PDF file here [Hidden Confidential Document](#) and uncover the flag within the metadata.

Hints 

1 2

35,266 users solved

95% Liked 

Submit Flag

- Step 1: Identify the Decoy Content The document contains several sections of *Lorem Ipsum* text and "Special Hidden Sections" that claim to hold secrets. However, Hint 1 explicitly states that the visible text is a decoy designed to distract investigators.

Title: *The Ultimate Guide to Flag Hunting*

Welcome to the challenge!

Don't worry, this might look like gibberish, but maybe there's something hidden somewhere? I spent so much time creating this PDF with care... or maybe not! 😊

Here's a Quick Story:

Once upon a time, in a land far, far away, there was a secret... But where could it be? Hidden deep within the document? Maybe the text holds clues?

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer posuere erat a ante venenatis dapibus posuere velit aliquet. Fusce dapibus, tellus ac cursus commodo, tortor mauris condimentum nibh, ut fermentum massa justo sit amet risus. Curabitur blandit tempus porttitor, consectetur adipiscing elit.

You thought this was important? Nah, it's just random text. Keep looking. Or maybe, just maybe, you're in the wrong place?

Aenean lacinia bibendum nulla sed consectetur.

Lorem ipsum dolor sit amet,

Special Hidden Section:

The author have done a great and good job

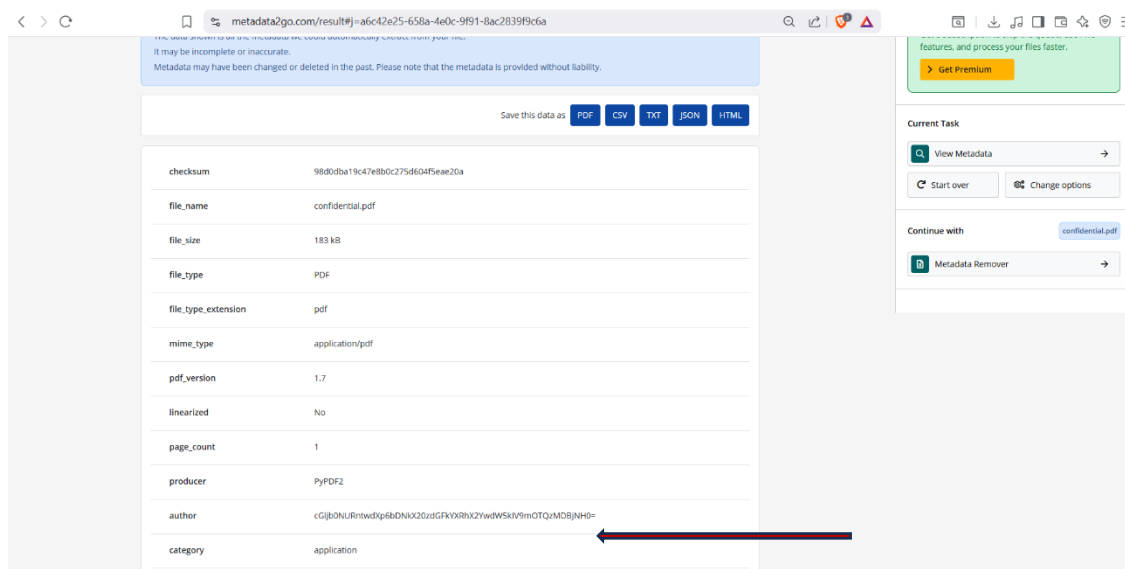
Don't bother trying to reveal the hidden text, it's just nonsense anyway. Even if you somehow manage to do it, all you'll get is:

No flag here. Nice try though!

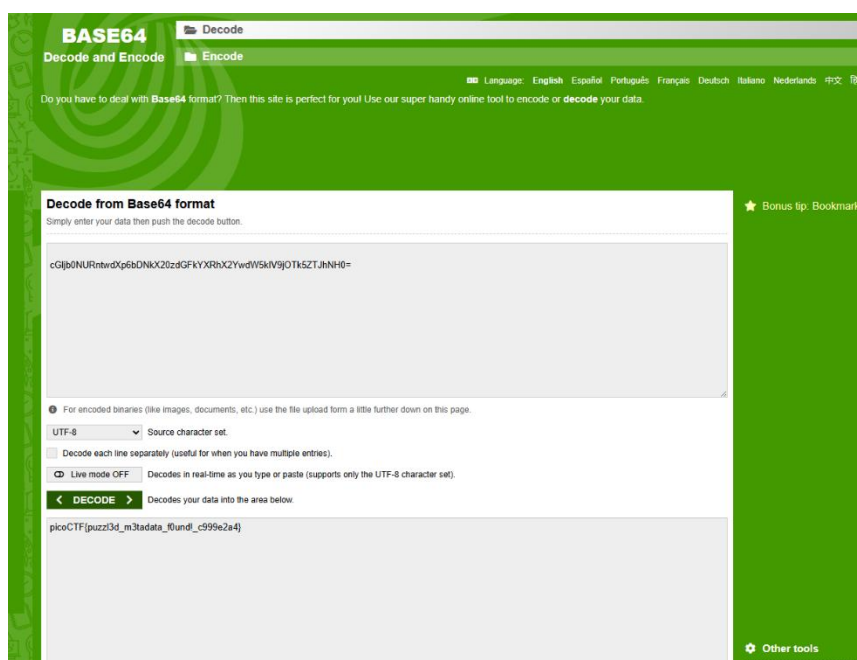
If you're still reading this, I'll tell you a secret: the answer might not be here after all... 😊

Good luck! You'll need it! 🍀

Metadata2Go.com is an online tool that lets users view and analyze hidden metadata in files such as images, videos, audio, and documents. It reveals information like creation dates, device or software used, file details, and sometimes GPS location, helping users check privacy and manage file data.



- **Step 2: Access the File Metadata** Following Hint 2 to "look beyond the surface," the file must be analyzed using a metadata viewer. Using **Metadata2Go.com**, the document's internal properties—which are not visible when simply reading the PDF—are revealed.
- **Step 3: Extract the Encoded String** Upon inspecting the metadata results, a long string of alphanumeric characters was discovered in the "author" field. The specific string found was cGlb0NURntwdXp6bDNkX20zdGFkYXRfOHVudZGxYFk5OWUyNH0=.



- **Base64** is a way to **convert binary data into text** using 64 ASCII characters, making it safe to transmit over text-only systems like email or JSON. It's encoding, not encryption, and is commonly used for files, images, or data embedding.
- **Step 4: Decode the Base64 Payload** Recognizing that Base64 is a common method for encoding data for transmission, the string was entered into a Base64 decoder. The trailing = sign is a classic indicator of Base64 padding.

- **Step 5: Reveal the Flag** The decoder converted the encoded string into plain text, revealing the final flag format.

2. Short Reflection

This challenge serves as a fundamental lesson in **digital forensics** and **information obfuscation**. It highlights that data is often stored in two layers: the *content* layer (what the user sees) and the *metadata* layer (the data about the data).

The primary takeaway is that "hiding in plain sight" often involves making the visible information so loud or confusing that the investigator forgets to check the container itself. By utilizing encoding (Base64) rather than encryption, the author created a hurdle that is easy to clear once the correct forensic tool—in this case, a metadata analyzer—is applied.